Application of Data Anonymization to Smartphone Apps Usage Logs

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***Abstract*—In recent years, the importance of data science has rapidly increased, along with opportunities for gaining new knowledge, such as through competitions. This research examines a method for anonymizing log data, using smartphone apps usage logs as an example, that maintains the value of the data while ensuring both anonymity and analyzability.**

We described a method for generating anonymized data by solving an assignment problem that that shuffles the usage apps, incorporating time series information, based on Pk-anonymization focused on suites of apps. And we explored three anonymizing methods.

Our verification using smartphone apps usage logs data for cluster analysis shows that all methods effectively replicate trend analysis. The method generating data to shuffle fully was found to be overprocessed, and considering time series information proved effective. Comparing two methods with and without Pk-annymization, it offered more robustness but slightly affect the accuracy of detailed values in cases of fewer logs. However without affecting the accuracy of cluster classification. Pk-anonymization was confirmed useful for maintaining confidentiality in third-party data provision or avoiding identification from niche app usage. Therefore, the method with Pk-anonymization is recommended for situations requiring anonymity, and one without Pk-anonymization for scenarios prioritizing analysis potential, including time series log analysis.

Future work are improving methods to better carry over time series information and to more accurately address cases with fewer logs, bringing detailed values closer to the original.

***Index Terms*—Data privacy, Big Data, Data security, Anonymization, Privacy Preserving Data Mining, Smartphone Apps, Log Data, Combinatorial optimization**

I. Introduction

Since Boyd and Crawford stated, ”The era of Big Data has begun” in 2012 [1], the significance of data science, including big data analysis, has been rapidly increasing in recent years. In Japan, the White Paper on Information and Communications [2] reveals that big data is being utilized across a wide range of fields. Amidst widespread application in industry and academia, opportunities for data science education and related competitions are on the rise. In recent years, several organizations are held data analysis competitions providing big data to foster ongoing discussion and techniques for AI and data analysis.

On the other hand, handling (and providing) large amounts of data raises issues of personal privacy protection. In the European Union (EU), the General Data Protection Regulation (GDPR) [3], which defines rules for processing personal data and requirements for transferring personal data from the European Economic Area (EEA) to third countries, was enacted in April 2016 to replace the Data Protection Directive 95 [4] and came into effect on May 25, 2018. Like this, in the desital society, regulations to protect personal information have been enacted around the world.

In Japan, the revised Personal Information Protection Law, enforced on May 30, 2017, introduced the concept of ”anonymously processed information.” This revision was motivated by the needs to generate an environment for the appropriate use of big data, including personal data. Moreover, the amendment enforced on April 1, 2020, further considers the importance of data, with a focus on adapting to the AI and Big Data era. New systems such as ”pseudonymously processed information” and ”personal-related information” have been established to maintain anonymity in personal information for data utilization, reflecting revisions in legal content to accommodate big data utilization and personal privacy protection.

The urgency and importance of privacy protection are increasing, especially with recent amendments that include logs (histories) detailing users' thought processes and behavior patterns. For example, logs can potentially identify individuals if combined with databases that can trace names from IDs, or when user numbers are low, raising privacy concerns. The Personal Information Protection Commission requires these logs to be treated as personal information, not personal-related information, if they have potential for easy association at either the provider or recipient end.

Regarding privacy protection technology, encryption and Secure Computing are generally used when the subject of protection is the accessors. When the subject is data, and especially when the analysis results are modified, differential privacy is employed. Database operations involve methods such as pseudonymization, shuffling (replacement or interchange), and information truncation to meet anonymity, such as k-anonymity proposed by Sweeney [5] and l-diversity by Ashwin et al. [6]. However, these methods face the challenge of reduced data utility due to deletion or rounding. To overcome such issues, Igarashi et al. [7] proposed Pk-anonymization, which maintains k-anonymity without reducing utility by adding noise to data and performing probabilistic operations.

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This method allows for creating valuable data while preserving anonymity.

Privacy protection research on big data has been active, and Jang [8] presented a scheme to overcome the problems of applying k-anonymity and l-diversity to big data anonymization. In terms of scalability, Zhang et al. [9] proposed an approach for multidimensional anonymization, while Mehta et al. [10] proposed an approach for data velocity. Addressing the challenges of access control, Gianluca Lax and Antonia Russo [11] proposed a blockchain-based solution for supporting anonymity and accountability. In the area of Privacy Preserving Data Mining (PPDM), Xu et al. [12] investigated various approaches to help protect sensitive information, and Anh T. [13] described a method to improve k-anonymity from the perspective of data mining. Also most recently Hewage et al. [14] discussed the pros and cons of existing PPDM and Privacy Preserving Data Stream Mining (PPDSM) methods. For log data, Pàmies-Estrems et al. [15] proposed a real-time anonymization solution for web search query logs, and Meng et al. [16] proposed a privacy-preserving query log sharing method. Fard and Wang [17] proposed a transaction anonymization technique based on clustering and generalization techniques to achieve k-anonymity privacy for web query logs. Portillo-Dominguez and Ayala-Rivera [18] proposed a sensitive data an automated approach was proposed to anonymize log streams in order to protect confidentiality and prevent third-party exposure or misuse.

In data related to smartphones, GPS data is carefully handled as location data; Hara et al. [19] described a method to anonymize user location data by generating dummies that behave like humans; Sei and Ohsuga [20] showed a method to anonymize location data by considering location error and the probability that each user is actually present in an anonymized area. Peng et al. [21] proposed a method to protect multi-user location correlation information under a strict privacy budget.

Among the related studies mentioned above, less known has been done on data related to smartphone applications. Access log data and smartphone app usage log data contain easily comparable information, such as detailed tracking of the websites visited or apps used, which can lead to individual identification. Therefore, anonymization methods are necessary when providing these data to third-party. Furthermore, it is essential to retain the usability and utility of the data as much as possible during provision.

Thus, this research examines a method for anonymizing smartphone app usage log data without losing its value. The technical domain targeted is database anonymization. Specifically, as described in section II, this involves generating anonymized data by applying Pk-anonymization and shuffling records. The data in focus is the smartphone app usage log data used by Yang and Shigeno [22]. The goal is to process the data for external provision or use, such as in a more protected form without losing its value, and to balance analyzability and privacy protection. In terms of analyzability, given the focus on big data and log data, the aim is to enable mass analysis, such as trend analysis of the entire dataset.

The envisioned use case for this method is for companies and organizations with log data to apply it and generate anonymized data that can be provided to third-party and research institutions. This method will enable the provision of data that could not be shared publicly due to confidentiality, contributing to further knowledge sharing, data science education, and the development of competitions.

II. Methodology

Smartphone app usage historical data for individual users can be ingredient to analyze app features and user behavior to use mobile devices. Even if data representing individuals are not held, the possibility of identifying an individual cannot be denied if one uses niche apps at abnormal times. Moreover, some developers hesitate about the clarity of the market for their apps. Therefore, we combine two methods to anonymize app usage log data for every unit time period.

(a) Anonymize focused on the suite of apps each user use as attribute value

(b) Shuffle by assigning users to suites of apps similar to they used and generate anonymized data

A. Notation

Let A be the set of target apps to analyze and $U(k)$ be the set of users who use at least one app in A during time period k. The suite of apps that user $i(\in U(k))$ during time period k is denoted by $S(k,i)(\subseteq A)$ and the set of suites of apps for time period k is denoted by $\mathscr{S}^k = \{S(k,i) \mid i \in U(k)\}$. Assume that each app belongs to exactly one of the categories such as communication, education, comics, games and so on. So, each category can be regarded as a subset of A and the set of categories, denoted by C is a partition of A.

B. Generating anonymized data

The first step of our process anonymizes the log data regarded the suite of apps of each user as user's attributes value. In this step, we employ Pk-anonymization method proposed by [7]. Here, each suite of apps in $\mathscr{S}^k$ is randomly changed to another suite of apps having the same most frequent category with the probability $1-\rho$, where the most frequent category of $S \in \mathscr{S}^k$ is defined by $c_S = \arg\max_{c \in C} |c \cap S|$. So, S can be changed to S' when $c_S = c_{S'}$. The Pk-anonymization method then reconstructs the data to estimate the original distribution of data from the statistical property. That is to say, we obtain the number of users using each $S \in \mathscr{S}^k$ with a few noises, from which the number of users using each app can be estimated.

By using the calculated number of users for each suite of apps, we generate anonymized data. The set of users $U(k)$ is kept, but the suite of the apps used by each user is changed to other similar suite in $\mathscr{S}^k$ with probability $(1-\gamma)$. The suites of apps randomly selected from $\mathscr{S}^k$ with probability γ does not change its users from the original data. Meanwhile, to shuffle the remaining suites of apps for the remaining users we solve an assignment problem. Let $\tilde{\mathscr{S}}^k$ be the set of target suites of apps of shuffling and $U(\tilde{\mathscr{S}}^k)$ be the set of users

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who used the suites of apps in $\tilde{\mathscr{S}}^k$ in time period k. Since the calculated number of users for each suite of apps by Pk-anonymization method may not be given by integer value, we obtain the number of users $\tilde{u}(k, S)$ for each suite of apps $S \in \tilde{\mathscr{S}}^k$ by applying Hamilton method of the apportionment problem so that the numbers are integers and their total is equivalent to the number of users in $U(\tilde{\mathscr{S}}^k)$. Then, each user $i(\in U(\tilde{\mathscr{S}}^k))$ is assigned to the suite of apps which is as similar as possible to $S(k, i)$, ensuring that $S(k, i) \neq S$ so that users with distinctive usage patterns are not identified. This assignment can be obtained by solving an assignment problem using the binary decision variable x_{iS} for whether user i is assigned to the suite of apps S or not. For each suite of apps $S \in \mathscr{S}^k \setminus \tilde{\mathscr{S}}^k$ that is not the target of the assignment problem, set $x_{iS} = 1$ when $S = S(k, i)$.

The assignment problem for time period k is represented by the following (1) to (3):

$$\text{minimize} \quad \sum_{i \in U(\tilde{\mathscr{S}}^k), S \in \tilde{\mathscr{S}}^k} \text{dist}(S(k,i), S) x_{iS} \tag{1}$$

subject to

$$\sum_{i \in U(\tilde{\mathscr{S}}^k)} x_{iS} = \tilde{u}(k, S), \quad \forall S \in \tilde{\mathscr{S}}^k \tag{2}$$

$$\sum_{S \in \tilde{\mathscr{S}}^k \setminus \{S(k,i)\}} x_{iS} = 1, \quad \forall i \in U(\tilde{\mathscr{S}}^k) \tag{3}$$

Equation (1) aims to minimize the distance between suites of apps before and after user assignment. Equation (2) represents the constraint ensuring that only the specified number $\tilde{u}(k, S)$ of users are assigned to each suite. Equation (3) ensures that each user is assigned to a different suite of apps than their original one.

C. Distance between suites of apps

We next define the distance between suite of apps $S_1, S_2 \subseteq A$, $\text{dist}(S_1, S_2)$, for the assignment to shaffle assigning users. The distance is given with the aid of demographic data of users using the apps in the whole periods, and information of categories, and developer of the apps. More precisely, we adopt the following information:

- $n(S)$: The number of apps in set S.
- A_i: Average age of users for the ith app in S. The average is calculated among users who use the app at least once during the whole period.
- G_i: Gender ratio (ratio of females) of the ith app in S. The average is calculated among users who use the app at least once during the whole period.
- C_j: Ratio of the number of apps belonging category j in S.
- D_j: Ratio of the number of apps developed by developer j in S.

A suite of apps $S \subseteq A$ is charactrized by two vectors

$$nAG(S) = (n(S), A_1, A_2, \ldots, A_{n(S)}, D_1, D_2, \ldots, D_{n(S)})$$

and

$$CD(S) = (C_1, C_2, \ldots, C_m, D_1, D_2, \ldots, D_n)$$

where m and n are the number of considering categories and the number of considering developers, respectively. To calculate the distance between suites of apps S_1 and S_2, we use two distinct measures: the Euclidean distance for the continuous valued vectors $nAG(S_1)$ and $nAG(S_2)$, and the cosine similarity for the categorical valued vectors $CD(S_1)$ and $CD(S_2)$. In this process, it is necessary to coincide the dimension of two vectors. Since the length of the vector $nAG(S)$ depends on the number of apps in S, we adjust the length of vectors, for calculate Euclidean distance between $nAG(S_1)$ and $nAG(S_2)$, as follows:

1) Determine $n_{\max} = \max\{n(S_1), n(S_2)\}$

2) If $n_{\max} > n(S_k)$ $(k = 1, 2)$, add values (padding) $A_{n(S_k)+1}, \ldots, A_{n_{\max}}$ by setting $\sum_{i=1,\ldots,n(S_k)} A_i / n(S_k)$, and $G_{n(S_k)+1}, \ldots, G_{n_{\max}}$ by setting $\sum_{i=1,\ldots,n(S_k)} G_i / n(S_k)$.

3) Reformulate $nAG(S_1)$ and $nAG(S_2)$ by sorting $(A_1, A_2 \ldots, A_{n_{\max}})$ and $(G_1, G_2, \ldots, G_{n_{\max}})$ in ascending order, respectively, to eliminate the impact of the order of apps.

4) Using the obtained vectors, calculate the Euclidean distance.

This process modifies for any imbalance in the number of apps in a suite without distorting the distribution of attribute values and generates vectors that can be applied to Euclidean distances. Realigning the value A_i and G_i in ascending order removes the impact of their order, allowing for distance calculation based solely on the distribution of attribute values. Additionally, normalizing the calculated Euclidean distance enables comparison between sets of different sizes and scales, so the normalized Euclidean distance $\text{normal_e_dist}(nAG(S_1), nAG(S_2))$ among all pairs of suites of apps in $\mathscr{S}^k$ is used in the computation of the comprehensive distance in the time period k.

Meanwhile, since the length of vector $CD(S)$ is $m + n$ for any $S \subseteq A$, we can obtain the cosine similarity, $\text{cos_sim}(CD(S1), CD(S2))$, which is based on the angle between the categorical valued vectors $CD(S_1)$ and $CD(S_2)$. Because the values of the vectors are nonnegative, the cosine similarity takes values between 0 and 1, which implies that we use it without normalization. In addition, we use it by transforming to distance by subtracting it from 1 so as that it closer to 0 represents the result with high similarity.

The final distance between S_1 and S_2 is given by

$$\begin{aligned}\text{dist}(S_1, S_2) = &\alpha \cdot \text{normal_e_dist}(nAG(S_1), nAG(S_2)) \\ &+ (1 - \alpha) \cdot (1 - \text{cos_sim}(CD(S_1), CD(S_2)))\end{aligned} \tag{4}$$

where $\alpha (0 \leq \alpha \leq 1)$ is a parameter that adjusts the importance of Euclidean distance and cosine similarity.

By the way, since our smartphone app logs are given as time series data, we use information of assignment solutions before kth time period when solving the assignment problem for time period k to generate more consistent anonymized data. When solving the assignment problem for the time period k, we define the weight to

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reflect the solution at past time period t, σ^{kt}, and introduce it to the distance between suites of apps. The past time periods reflected in the distances, based on the domain knowledge of smartphone app logs and user's life patterns, are selected as following four kinds:

(i) 1 hour ago (considering hourly updates within apps and lifestyle habits)

(ii) 8 hours ago (considering commuting or school hours. Also, according to "App Market White Paper 2021" [23], "Users launch the same app an average of 3.5 times per day")

(iii) 1 day ago (considering the daily update timing of apps and circadian rhythm-based habits)

(iv) 1 week ago (considering lifestyle habits and app's weekly update events)

Let $T(k)$ be a set of four time periods considered for computing time period k. In order to reflect to the current assignment problem form optimal solutions in the past time periods, if a user i with $S_1 = S(t,i)$ is assigned to S_2 in the optimal solution for time period $t \in T(k)$, the distance between S_1 and S_2 is reduced to make easier to assign such a pair of user and suit of apps. Thus, we modify the distance between suites of apps S_1 and S_2 by using optimal solutions for past assignment problems and weight between time periods t and k, σ^{kt}, as

$$\mathrm{T_dist}^k(S_1, S_2) = \sum_{t \in T(k)} \left((1 - \sigma^{kt}) \sum_{\substack{i \in U(t) \\ S(t,i) = S_1}} \tilde{x}^t_{iS_2} \right) \mathrm{dist}(S_1, S_2)$$

where $\tilde{x}^t_{iS}$ is an optimal solution for the assignment problem for time period t.

The weight σ^{kt} which represents the reflection from the time period t to the time period k estimates degree of closeness between $\mathscr{S}^t$ and $\mathscr{S}^k$. The closeness also evaluated by distance of suites of all apps in $\mathscr{S}^t$ and $\mathscr{S}^k$, respectively. That is to say, it is given by $\mathrm{dist}(\bigcup_{S \in \mathscr{S}^k} S, \bigcup_{S \in \mathscr{S}^t} S)$ provided that the averages and ratios in vectors $nAG(S)$ and $CD(S)$ are weighted by how many suites include an app. By defining $S^k = \bigcup_{S \in \mathscr{S}^k} S$, the weight σ^{kt} is given by

$$\sigma^{kt} = \frac{\mathrm{dist}(S^k, S^t)}{\sum_{t' \in T(k)} \mathrm{dist}(S^k, S^{t'})} \cdot \frac{1}{\sum_{i \in U(\mathscr{S}^k), S \in \mathscr{S}^k} \tilde{x}^t_{iS}}.$$

Fig. 1 displays the process of computing distances for the assignment problems.

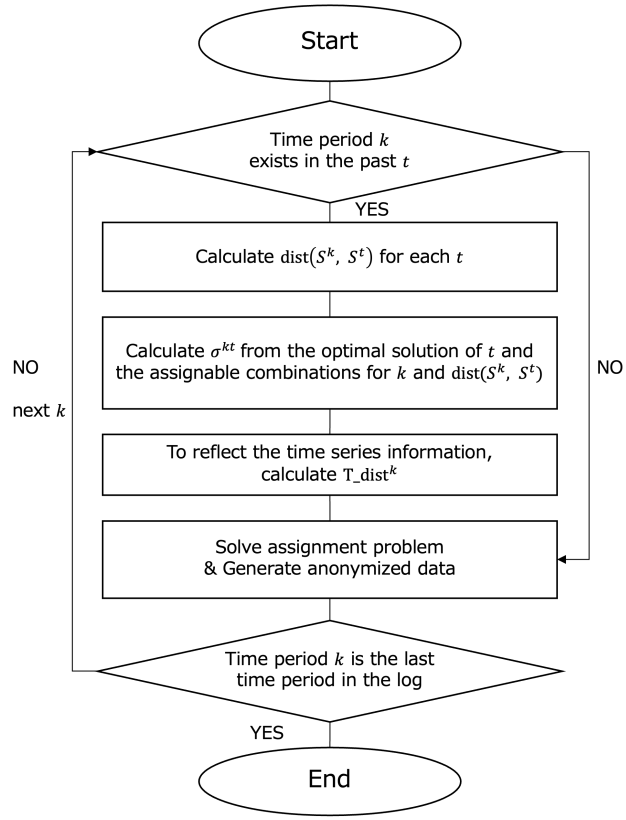

Fig. 1: Process of computing distances for the assignment problems

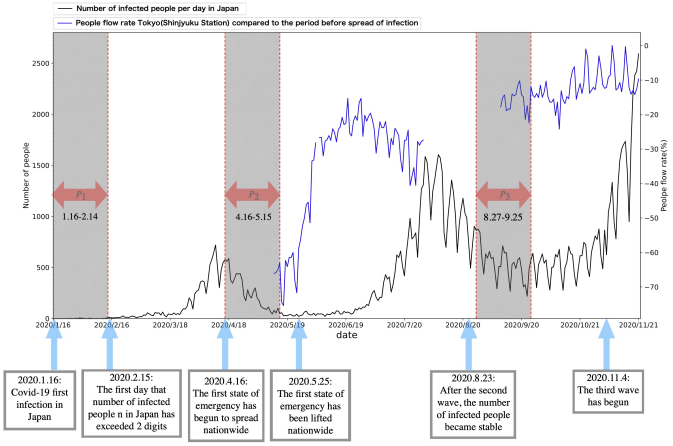

Figure 1: **The daily number of infections nationwide and the flow rate of Tokyo**

Fig. 2: Analysis target time periods (cited from Yang and Shigeno [22])

III. Results and Discussion

The anonymized data generated will be used to replicate the analysis of Yang and Shigeno [22], and its usefulness will be evaluated based on the degree of reproducibility of the analysis results.

Yang and Shigeno [22] observed and analyzed the impact of the pandemic on people's lives using changes in smartphone apps usage patterns. They observed the usage of 91 top-ranking apps across six categories, recognized to be significantly affected by the pandemic of COVID19, over a total of 66 days in three periods: Pre-pandemic (P1), Self-restraint period (P2), and Post-restraint period (P3), each hour from 0 to 23 hours. The target users were those who launched the target apps at least once in all periods P1, P2, and P3. Fig. 2 detail the analysis target time periods.

A. Verification Method

On the basis of the method described in section II, we compare the usefulness of each method by generating and validating anonymized data using three methods categorized in the following steps.

1) "Pk-AS (Pk-anonymized Assignment Shuffle)": Generate completely shuffled anonymized data by solving the assignment problem with distance measured by dist. I.e., anonymized data are generated without using time series weighting and setting γ to 0 after performing Pk-anonymization. "Pk-AS" is a method that deviates from the original datas as it changes the number of users using each app due to Pk-anonymization and significantly changes pairs of users and suites of apps due to complete shuffling.
2) "Pk-TAS (Pk-anonymized Time series Assignment Shuffle)": Set an appropriate value γ after Pk-anonymization and solve the assignment problem by measuring distance with T_dist

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considering time series, with a probability γ keeping the same as the original datas. "Pk-TAS" is a method that focuses on conservativeness and analyzability by keeping the same as the original datas probabilistically while including time series, as compared to "Pk-AS".

3) "TAS (Time series Assignment Shuffle)": Solve the assignment problem by measuring distance with T_dist considering time series without performing Pk-anonymization, and setting γ. "TAS" focuses more on analyzability, as it does not change the number of users using each app without Pk-anonymization, only changing pairs of users and suites of apps. Hence it is a method with lower processing but higher analyzability.

Common parameters for each method include setting the time period k for creating anonymized data to every 15 minutes and the parameter α related to dist calculated in (4) to $\alpha = 0.5$. For the calculation of dist(S^k, S^t), to improve calculation efficiency and correctly calculate distance based on the number and frequency of app usage, suites of apps $S(k,i)$ from $\mathscr{S}^k$ and $S(t,i)$ from $\mathscr{S}^t$ were randomly sampled at a 50% sampling rate. Regarding the selection of the value of the maintenance probability ρ for Pk-anonymization, we followed the approach used in PWSCUP2016[1] [24], [25]. The ut-jaccard coefficient treated in this research is used as a measure of the degree of anonymization, defined as:

$$utj = \frac{1}{|K| \cdot |U|} \sum_{k \in K} \sum_{i \in U} \left(1 - \frac{|S(k,i) \cap S'(k,i)|}{|S(k,i) \cup S'(k,i)|} \right) \tag{5}$$

where U represents the set of users and K represents the set of time periods during the target period, $S^{'}(k,i)$ represents suites of apps of user i after Pk-anonymization in time period k. This coefficient reflects the extent of changes in a suite of apps before and after anonymization. According to [24], [25], a ut-jaccard value above 0.7 is regarded as to overprocessed. Our results in Table. I indicate that setting $\rho \geq 0.3$ effectively prevents overprocessing, with $\rho = 0.3$ allowing for maximal anonymization without loss of data value. Hence, for Pk-anonymization, we employ $\rho = 0.3$.

Similar to probability ρ, based on the overprocessing criteria of [24], [25], if not completely shuffling, γ is set to 0.3.

The utj of (5) is also measured to determine how much the anonymized data deviates from the original one, and it is used as an indicator of anonymity. Next, the cluster analysis [22] is replicated. Let $F_P(i,h)$ be the number of days user i used the target apps during time width h in period P, and users are classified into 9 clusters by k-means++ based on $F_{P1}(i,h) - F_{P2}(i,h)$ and $F_{P2}(i,h) - F_{P3}(i,h)$. The characteristics of each cluster's change are named $C^{+,+}$, $C^{+,0}$, $C^{+,-}$, $C^{0,+}$, $C^{0,0}$, $C^{0,-}$, $C^{-,+}$, $C^{-,0}$, $C^{-,-}$. For example, users who tend to have a $+$ change from P1 to P2 and a $-$ change from P2 to P3 belong to the $C^{+,-}$ cluster.

Then, we compare the average number of usage days $F_P(h)$ among users in each cluster between the original data and generated anonymized data. The evaluation of anonymity and analyzability of anonymized data is conducted from the points as the following:

- The values of utj as anonymity.
- The correlation coefficient for $F_P(h)$ by the original data and anonymized data in the same cluster as reproducibility of the trend of data.
- The mean absolute percentage error of $F_P(h)$ between the original data and anonymized data as reproducibility of the detailed value.
- The differences of set of users belonging to each cluster and user attribute ratios (age, gender) between the original data and anonymized data for each cluster as the clustering reproducibility.

[1]PWSCUP is an annual competition in Japan focusing on anonymizing a common dataset and evaluating the re-identification performance of the processed data. PWSCUP2016 [24], [25] employs the Online Retail Data Set [26], which contains a year's purchase history from a UK-based online retail site, similar to the historical log data context of our research.

TABLE I: Average utj for probability ρ

ρ	.1	.2	.3	.4	.5	.6	.7	.8	.9
utj	.90	.80	.70	.60	.50	.40	.31	.20	.10

B. Anonymity

Table. II compares the utj values calculated using (5) for each period. From this table, it can be seen that "Pk-AS" is overprocessed according to the criteria [24], [25], as the value is consistently greater than 0.7 across any periods. Both "Pk-TAS" and "TAS" were within an appropriate range of processing in all periods, and "Pk-TAS", which performs Pk-anonymization, had a larger utj and can be considered more robust.

C. Result for reproducibility of trend analysis

As part of the verification of analyzability, we evaluated the degree of trend reproducibility using the correlation coefficient derived from the average number of usage days $F_P(h)$ in the same cluster for the original data and each anonymized data. Table. III shows the correlation coefficients for each method across periods and clusters. It can be seen that every correlation coefficient takes more than 0.9 for all clusters in each period in all anonymized data, which implies that all methods accurately captured the overall usage trends. Fig. 3 is a line graph showing the hourly transitions of the average number of usage days per cluster for the original data. Similarly, Fig. 4 to Fig. 6 are graphs for the cases of "Pk-AS", "Pk-TAS", and "TAS" anonymized data, respectively. A comparison of these four graphs also shows that all anonymized data accurately captured the overall usage trends. However, there were differences in the specific values on the vertical axis for each variation.

D. Result for reproducibility of Detail Values

Let $F_P(c,h)$ and $\hat{F}_p(c,h)$ be the average numbers of usage days in cluster c for a time width h in period p in the original data and in the anonymized data, respectively. The Mean Absolute Percentage Error (MAPE) for each period and each time width is represented by (6), and each period MAPE by (7), each time width MAPE by (8).

$$\mathrm{MAPE_{P,h}} = \frac{1}{9} \sum_{\mathrm{c}} \left| \frac{\mathrm{F_P(c,h)} - \hat{\mathrm{F}}_{\mathrm{P}}\mathrm{(c,h)}}{\mathrm{F_P(c,h)}} \right| \times 100\% \tag{6}$$

TABLE II: Comparison of utj for each period

Method	P1	P2	P3
Pk-AS	0.769	0.769	0.772
Pk-TAS	0.321	0.322	0.319
TAS	0.292	0.296	0.290

TABLE III: Correlation coefficients for each periods and clusters

Method	Period	Cluster								
		$C^{+,+}$	$C^{+,0}$	$C^{+,-}$	$C^{0,+}$	$C^{0,0}$	$C^{0,-}$	$C^{-,+}$	$C^{-,0}$	$C^{-,-}$
Pk-AS	P1	0.954	0.982	0.987	0.991	0.995	0.994	0.997	0.997	0.997
	P2	0.991	0.997	0.997	0.994	0.995	0.998	0.989	0.992	0.994
	P3	0.998	0.996	0.987	0.997	0.995	0.981	0.998	0.993	0.939
Pk-TAS	P1	0.966	0.986	0.991	0.994	0.996	0.994	0.998	0.999	0.996
	P2	0.992	0.998	0.998	0.996	0.997	0.998	0.993	0.994	0.997
	P3	0.998	0.996	0.991	0.998	0.995	0.979	0.998	0.995	0.955
TAS	P1	0.972	0.989	0.994	0.995	0.997	0.997	0.999	0.999	0.998
	P2	0.996	0.999	0.999	0.996	0.997	0.998	0.996	0.995	0.997
	P3	0.999	0.998	0.995	0.998	0.996	0.990	0.999	0.996	0.945

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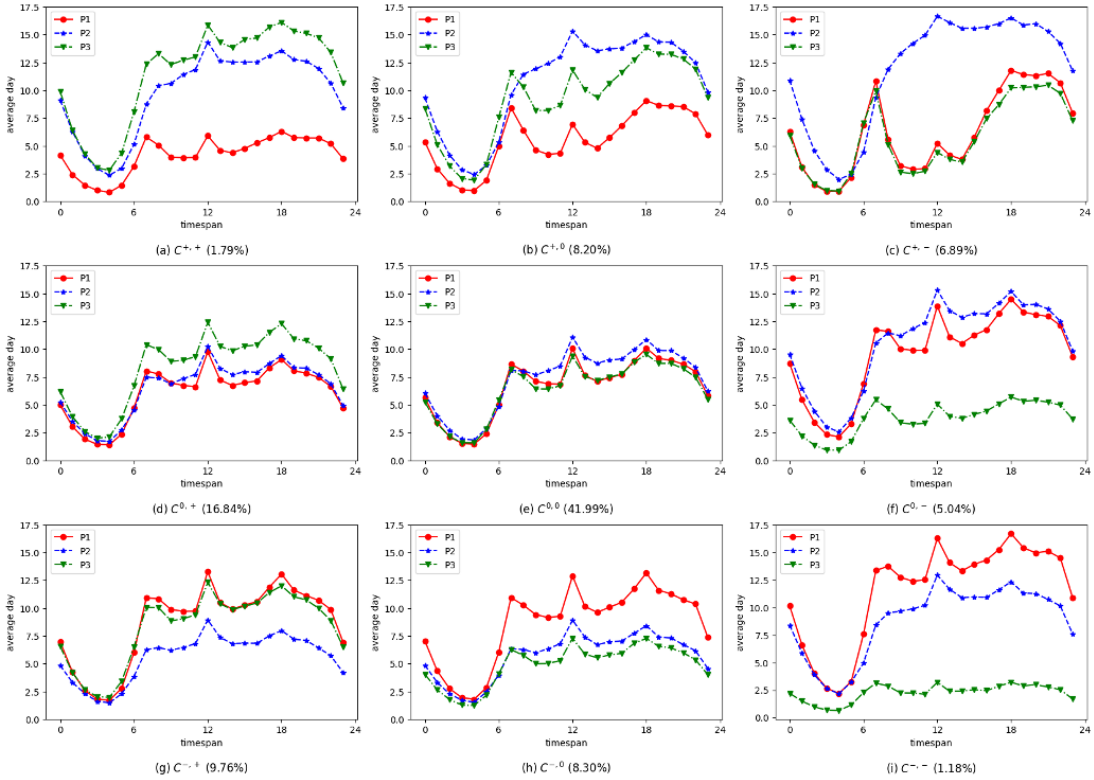
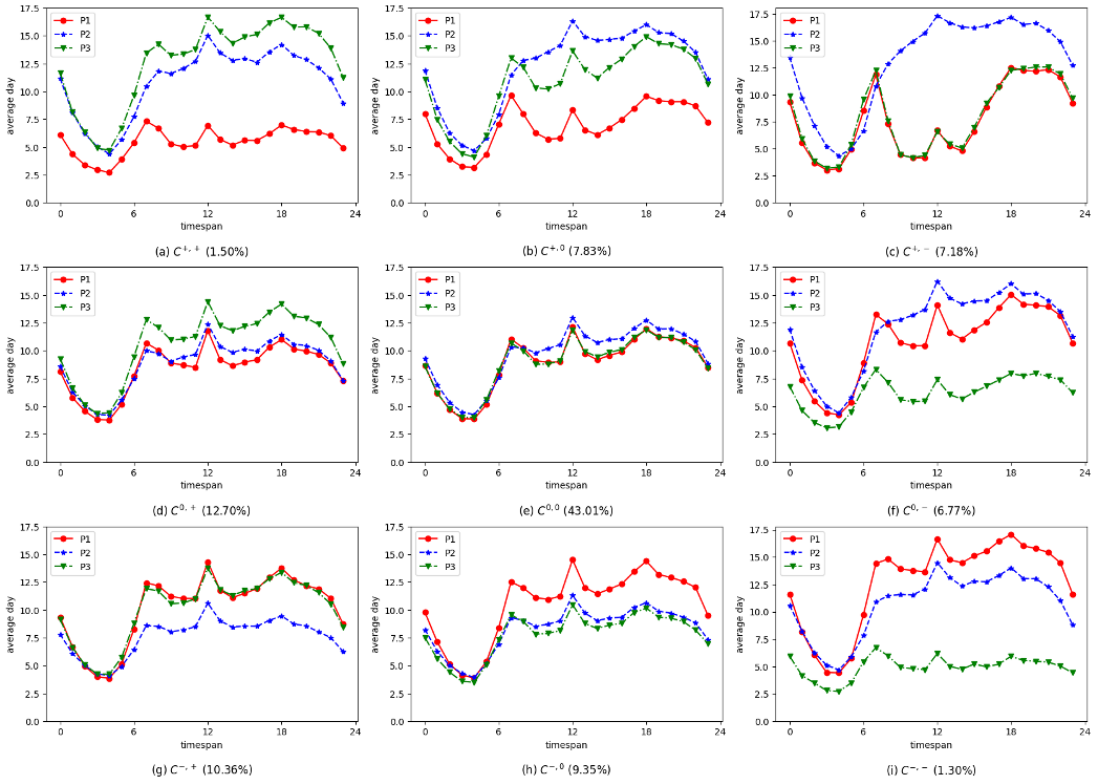


(a) $C^{+,+}$ (1.79%) (b) $C^{+,0}$ (8.20%) (c) $C^{+,-}$ (6.89%)

(d) $C^{0,+}$ (16.84%) (e) $C^{0,0}$ (41.99%) (f) $C^{0,-}$ (5.04%)

(g) $C^{-,+}$ (9.76%) (h) $C^{-,0}$ (8.30%) (i) $C^{-,-}$ (1.18%)

Fig. 3: Average number of usage days per cluster in the original data



(a) $C^{+,+}$ (1.50%) (b) $C^{+,0}$ (7.83%) (c) $C^{+,-}$ (7.18%)

(d) $C^{0,+}$ (12.70%) (e) $C^{0,0}$ (43.01%) (f) $C^{0,-}$ (6.77%)

(g) $C^{-,+}$ (10.36%) (h) $C^{-,0}$ (9.35%) (i) $C^{-,-}$ (1.30%)

Fig. 4: Average number of usage days per cluster in the anonymized data by Pk-AS

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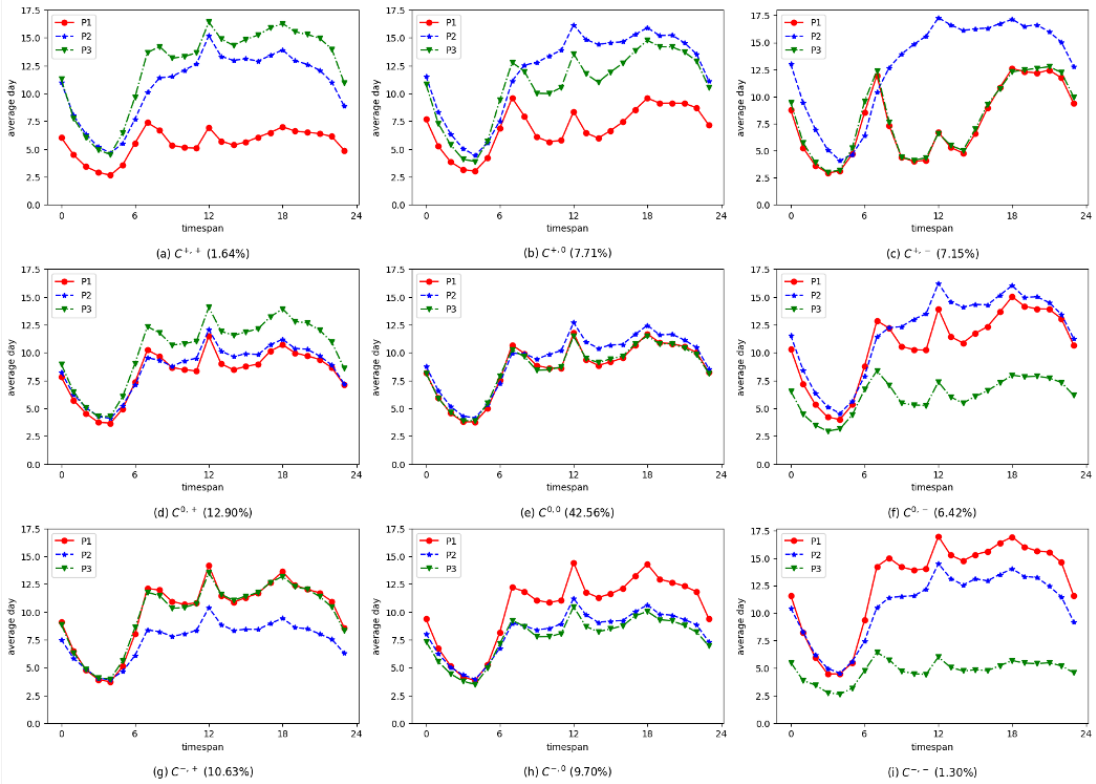

Fig. 5: Average number of usage days per cluster in the anonymized data by Pk-TAS

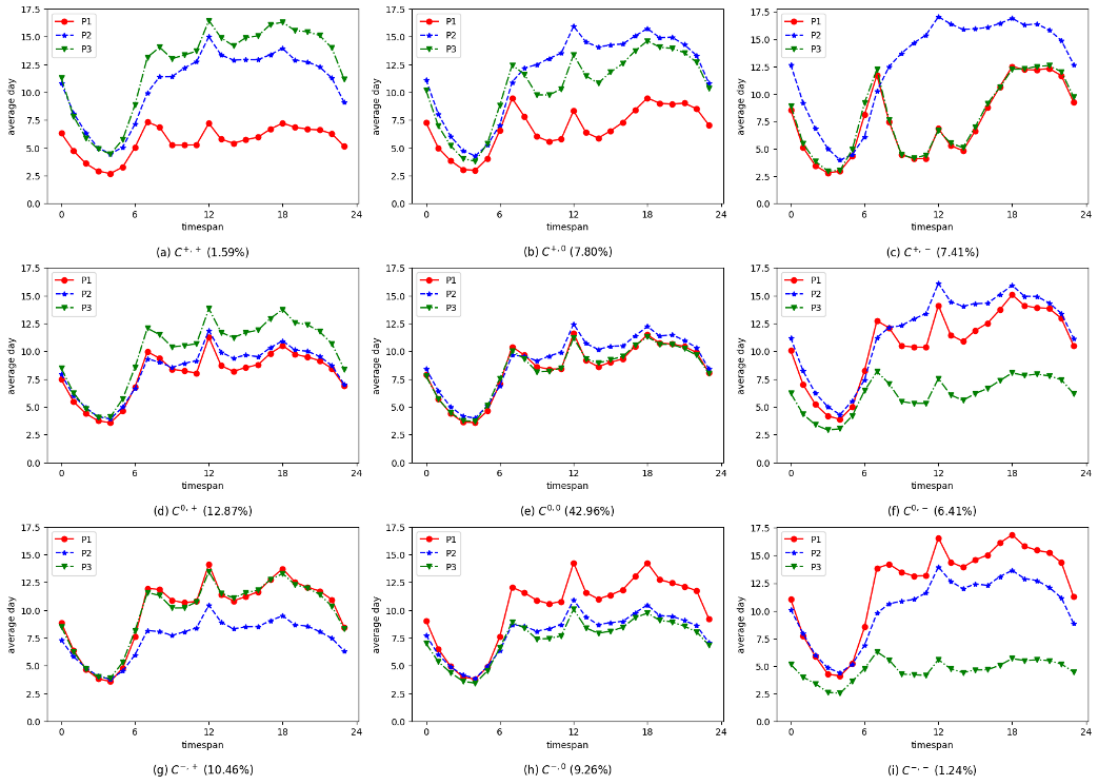

Fig. 6: Average number of usage days per cluster in the anonymized data by TAS

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TABLE IV: Comparison of $\mathrm{MAPE_P}$ and percentage of log counts in the original data

Method	P1	P2	P3
Pk-AS	40.44%	35.80%	59.24%
Pk-TAS	38.18%	33.58%	55.81%
TAS	35.37%	30.27%	52.34%
Percentage of log counts	32.76%	36.76%	30.49%

$$\mathrm{MAPE_P} = \frac{1}{24}\sum_{\mathrm{h}} \mathrm{MAPE_{P,h}} \tag{7}$$

$$\mathrm{MAPE_h} = \frac{1}{3}\sum_{\mathrm{P}} \mathrm{MAPE_{P,h}} \tag{8}$$

Table. IV compares $\mathrm{MAPE_P}$ and "percentage of log counts" for each anonymized data, and Figure. 7 shows the "$\mathrm{MAPE_h}$(left axis)" for each anonymized data and the "percentage of log counts(right axis)" in the original data.

From Table. IV shows that the number of logs is smaller in the order of P3, P1, and P2. In addition, there was no significant difference in $\mathrm{MAPE_P}$ when comparing the anonymized data over a specific time period, but there was a difference in the order of P3, P1, and P2 for all anonymized data, with larger values. This is due to the fact that fewer logs resulted in a smaller variety of suites of apps, and consequently a lower likelihood of being assigned to "more similar" suites of apps.

Comparing by time width as shown in Figure. 7, the difference in $\mathrm{MAPE_h}$ is larger during the late night and early morning hours, while the $\mathrm{MAPE_h}$ during the daytime hours, which is generally the center of app analysis, is kept at around 20%. The reason why $\mathrm{MAPE_h}$ was larger during late night and early morning hours than during the daytime is that, similar to $\mathrm{MAPE_P}$, these hours were less likely to be assigned to "more similar" suites of apps due to fewer logs. Furthermore, the errors are larger in the order of "TAS", "Pk-TAS", and "Pk-AS" during these time periods, suggesting that the effect of fewer logs is more pronounced for "Pk-AS", "Pk-TAS", and "TAS", respectively.

E. Result of reproducibility of the Clustering

Lastly, we evaluated clustering reproducibility. The age and gender ratios within each cluster in the original data and the anonymized data were compared. Fig. 8 and Fig. 9 show these as 100% stacked bar charts, respectively. Additionally, the correlation coefficients for each cluster and attribute ratios are noted in the captions of each figure. As a result, from the overall shape and correlation coefficients in Fig. 8a to Fig. 8d and Fig. 9a to Fig. 9d, it can be seen that the cluster attributes of the original data and each anonymized data generally matched.

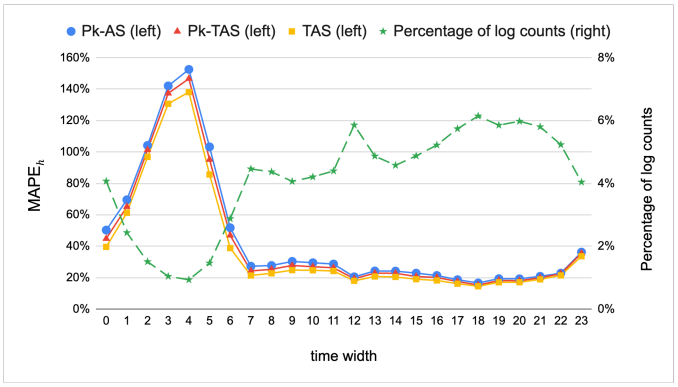
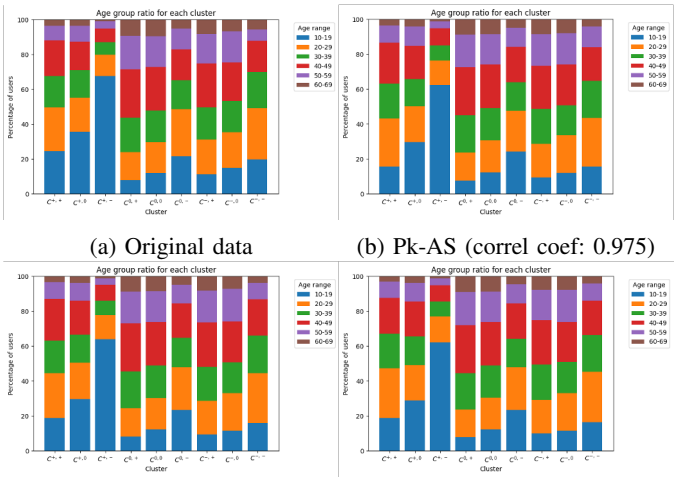


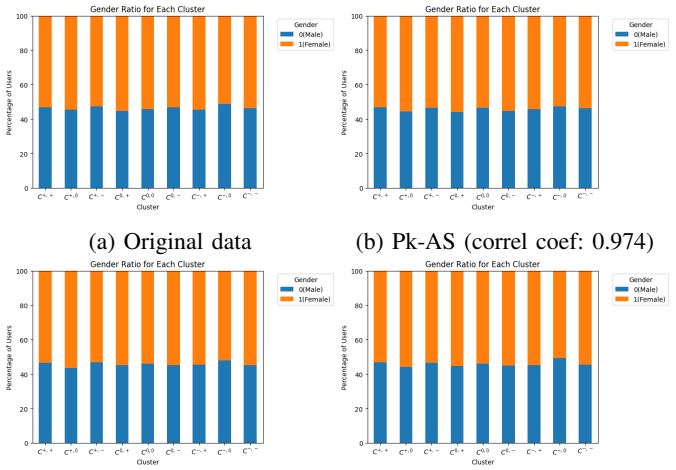
Fig. 7: $\mathrm{MAPE_h}$ for each time width and percentage of log counts



(a) Original data (b) Pk-AS (correl coef: 0.975)

(c) Pk-TAS (correl coef: 0.983) (d) TAS (correl coef: 0.982)

Fig. 8: Cluster age ratios in the original data and each anonymized data



(a) Original data (b) Pk-AS (correl coef: 0.974)

(c) Pk-TAS (correl coef: 0.982) (d) TAS (correl coef: 0.983)

Fig. 9: Cluster gender ratios in the original data and each anonymized data

Next, the focus is on the difference of clusters to which users were classified between the original data and each anonymized data. Table. V shows the ratios of users who were classified to a cluster with the same label with the original data and anonymized data. It can be seen that such a ratio was highest for "TAS", followed by "Pk-TAS" and "Pk-AS", with "TAS" and "Pk-TAS" being roughly the same.

To compare the ratio of users changed clusters from each cluster in the original data, we evaluated the ratio of users who were classified to a cluster having different label corresponding to both periods P1 to P2 and P2 to P3, as well as those whose cluster's label underwent a two-step change in one of the periods, in the anonymized data. A two-step change refers to users whose cluster was labaled as $+$ in the original data but became $-$ in the anonymized data or vice versa.

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Table. VI lists the obtained such ratios, called ratios of change, for each cluster. As a result, the ratios of change was lower in "Pk-TAS" and "TAS" than in "Pk-AS", which was better, and which of "Pk-TAS" or "TAS" was better varies depending on the focused cluster, indicating roughly the same level of clustering reproducibility.

F. Summary of Results

As indicated in subsections III-C and III-E, the trend reproducibility and the attribute (age and gender) ratio within each cluster for each anonymized data were very similar to the original data, indicating that analyses at a larger granularity, such as trend and attribute analysis, could be reproduced accurately. However, the evaluation of utj in subsection III-B, it was clear that "Pk-AS" was overprocessed. Comparing the ratio of users classified difference clusters in "Pk-AS" and "Pk-TAS" in subsection III-E, it was evident that considering time series was effective. Due to MAPE discussed in subsection III-D, during daytime hours, which are typically the focus of app analysis, generated anonymized data managed to keep the error around approximately 20%. It was observed that the affect of fewer logs during late night and early morning hours was more pronounced in "Pk-AS", "Pk-TAS", and "TAS", in that order. This implies that while "Pk-TAS" becomes more robust with Pk-anonymization, it slightly affected the detailed numerical values in cases (or time periods) with fewer logs, but did not change the overall reproducibility of clustering classification. Therefore, it was considered appropriate to use "Pk-TAS" and "TAS" selectively, taking into account both anonymity and analyzability.

Here, the advantages and demands of Pk-anonymization when considering real-world applications are discussed as follows:

1) When providing data to third-party as public datasets, if there is a request from the owning owner and/or partner companies to "delete certain app data usage information", it usually requires rounding off the data or completely deleting the relevant app information. However, Pk-anonymization allows anonymizing information (types, frequency of use, and number of users) focused on apps, which does not need for such processing potentially and enable the provision of a broaderrange of data.
2) Without Pk-anonymization, the original app information remains, so there is a possibility of tracing back to the users who access niche apps. Pk-anonymization can address such issue.

These points are applicable not only to app usage data but also to access log data and general log data. As a result, we conclude that, when there is a request mentioned above or when anonymity is a concern, it is advisable to use "Pk-TAS". Otherwise, when it's also important to consider the analyzability in cases with fewer logs, "TAS" should be preferred.

TABLE V: Comparison of label match user rate with the original data

Method	Label match user rate
Pk-AS	64.234%
Pk-TAS	68.302%
TAS	68.321%

TABLE VI: Ratio of changes for each cluster from the original data

Cluster	Pk-AS	Pk-TAS	TAS
$C^{+,+}$	7.500%	5.625%	6.458%
$C^{+,0}$	3.371%	2.187%	2.870%
$C^{+,-}$	12.779%	11.963%	12.289%
$C^{0,+}$	5.573%	3.923%	3.879%
$C^{0,0}$	4.986%	3.789%	3.699%
$C^{0,-}$	2.461%	1.939%	1.790%
$C^{-,+}$	12.500%	8.742%	9.663%
$C^{-,0}$	4.897%	3.158%	2.517%
$C^{-,-}$	15.016%	13.738%	14.058%

IV. Conclusion

This research focused on privacy protection, a core aspect of big data utilization, examined anonymization methods that balance anonymity and analyzability without losing the value of log data. Using smartphone apps usage logs as an example, we described a method generating anonymized data by solving an assignment problem that minimizes the defined distance between suites of apps, incorporating time series information, based on Pk-anonymization focused on app usage sets.

Moreover, we investigated three types of the method: "Pk-AS", which fully shuffles data; "Pk-TAS", which considers temporal information in assignment; and "TAS", which considers temporal information without Pk-anonymization. For verification about these, we replicated cluster analysis using data from Yang and Shigeno [22] and evaluated the degree of anonymization of the anonymized data, as well as the reproducibility of trend analysis, analytical detail values, and clustering. It is advisable to use "Pk-TAS" in situations where anonymity is a concern and "TAS" where analysis feasibility in cases with fewer logs is also a consideration.

Future work includes improving methods to better carry over time series information and enhancing methods to more accurately address cases with fewer logs, bringing detailed values closer to the original.

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